

Magnetic Fields & Biot-Savart Law

Magnetic Field Fundamentals

- A static charge produces only an electric field, whereas a moving charge produces both electric and magnetic fields.
- A current-carrying conductor produces only a magnetic field.

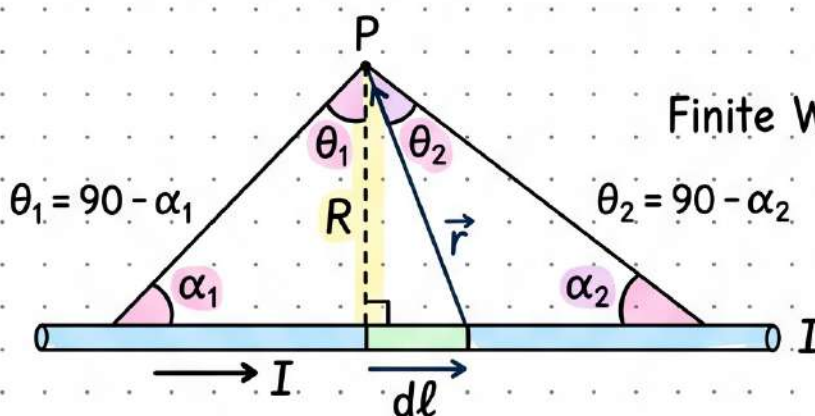
Biot-Savart's Law

The magnetic induction dB produced by a current element $I d\ell$ at a distance r is given by: $dB = \frac{\mu_0 \mu_r}{4\pi} \frac{I d\ell \sin \theta}{r^2}$.

In vector form: $d\vec{B} = \frac{\mu_0 \mu_r}{4\pi} \frac{I (d\vec{\ell} \times \vec{r})}{r^3}$.

μ_0 is the permeability of free space ($\mu_0 = 4\pi \times 10^{-7} \text{ H m}^{-1}$).

Straight Current Conductor



$$\begin{aligned} \text{Finite Wire: } B &= \frac{\mu_0 I}{4\pi R} (\cos \theta_1 + \cos \theta_2) \\ &= \frac{\mu_0 I}{4\pi R} (\sin \alpha_1 + \sin \alpha_2) \end{aligned}$$

↳ Infinitely Long Wire: $B = \frac{\mu_0 I}{2\pi R}$ (where $\alpha_1 = 90^\circ$, $\alpha_2 = 90^\circ$).

↳ Semi-Infinite Wire: $B = \frac{\mu_0 I}{4\pi R}$ (where $\alpha_1 = 0^\circ$, $\alpha_2 = 90^\circ$).

Coils, Solenoids & Ampere's Law

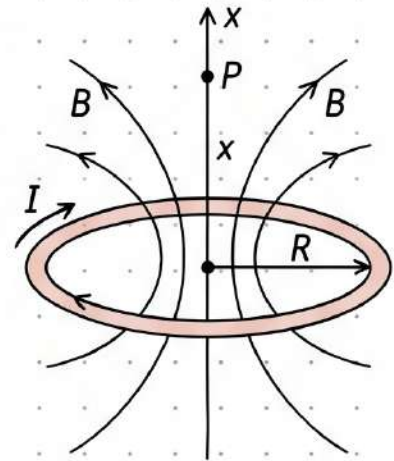
Flat Circular Coil (Current I , Radius R , Turns N)

At its centre: $B = \frac{\mu_0 NI}{2R}$

On the axis: $B = \frac{\mu_0 NI R^2}{2(x^2 + R^2)^{3/2}}$

(where x is the distance from the centre)

Circular Arc at centre: $B = \frac{\mu_0 I \theta}{4\pi R}$



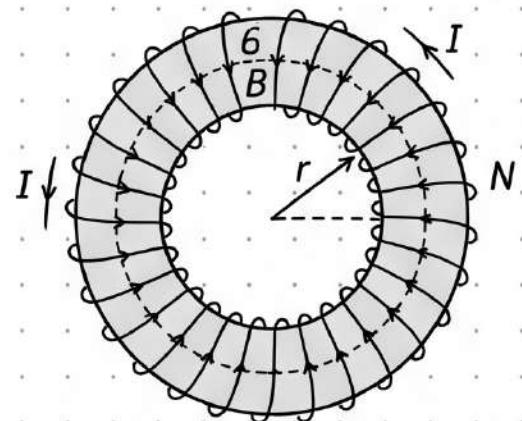
Ampere's Circuital Law

$$\oint \vec{B} \cdot d\vec{\ell} = \mu_0 \sum I, \text{ where } \sum I \text{ is the algebraic sum of all the enclosed current.}$$

Solenoid and Toroid

Solenoid: $B = \mu_0 n I$, where n is the number of turns per meter.

Toroid: $B = \mu_0 n I$, where $n = \frac{N}{2\pi R}$ and N is total turns.



Solid Cylindrical Conductor (Radius R)

Outside ($r \geq R$): $B = \frac{\mu_0 I}{2\pi r}$

Inside ($r < R$): $B = \frac{\mu_0 I r}{2\pi R^2}$

Moving Charges & Lorentz Force

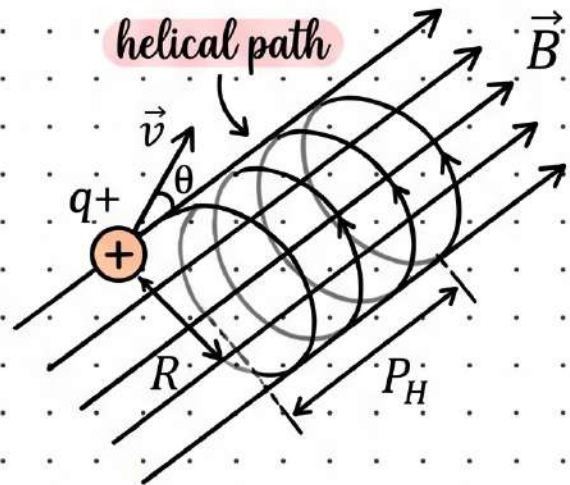
Helical Motion ($\vec{v}$ at angle θ to $\vec{B}$)

Radius : $R = \frac{mv \sin \theta}{qB}$

↗ radius

Pitch : $P_H = \frac{2\pi m v \cos \theta}{qB}$

↙ pitch



Lorentz Force

The net electromagnetic force is

$$\vec{F} = q\vec{E} + q(\vec{v} \times \vec{B})$$

↘ net force

Velocity Selector

- ↗ velocity selector
- When $\vec{v} \perp \vec{B}$ and $\vec{v} \perp \vec{E}$, the particle moves **undeflected** and **undeviated** with uniform speed if $v = \frac{E}{B}$.
 - If $\vec{v} \parallel \vec{B}$ and $\vec{v} \parallel \vec{E}$, motion is **uniformly accelerated** in a straight line as magnetic force is zero.

Magnetic Forces & Torque

Force on Current Carrying Wire

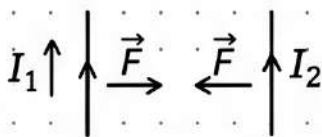
- **Straight Wire:** $\vec{F} = I(\vec{L} \times \vec{B})$
 - I = current
 - $\vec{L}$ = length vector
 - $\vec{B}$ = magnetic field vector

- **Random Shaped Conductor**

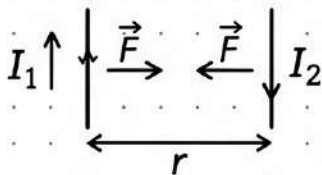
↳ Force is equivalent to the force on a straight wire joining the start and end points in a uniform magnetic field.

Interaction Between Two Parallel Wires

- **Force per unit length:** $F = \frac{\mu_0}{2\pi} \frac{I_1 I_2}{r}$



Attractive Force: Currents are parallel

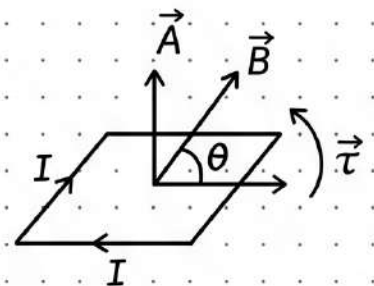


Repulsive Force: Currents are anti-parallel

The force is attractive if currents are parallel, and repulsive if currents are anti-parallel.

Torque on a Closed Current Loop

- **Torque:** $\vec{\tau} = NI\vec{A} \times \vec{B} = \vec{M} \times \vec{B} = BIN\sin\theta$
- **Magnetic Moment:** $\vec{M} = NI\vec{A}$



Magnetic Moment of Revolving Charge

- If a charge q is revolving at an angular velocity ω , its magnetic moment is $M = \frac{1}{2} q\omega R^2$.